Module 2: Git Assignment - 4

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Objective:

The main objective of Assignment 4 is to understand Git branching, merging, and synchronizing code across multiple development branches.

It simulates a situation where multiple developers work independently on the same project, and their changes eventually need to be integrated.

Tasks To Perform:

1. Put main.txt on main branch, stage and commit
2. Create 3 branches: public1, public2 and private3.
3. Put public1.txt on public 1 branch, stage and commit
4. Merge public 1 on main branch
5. Merge public 2 on main branch
6. Edit main.txt on private branch, stage and commit
7. Now update branch public 1 and public 2 with new main code in private
8. Also update new main code on main
9. Finally update all the code on the private branch

Solution Steps:

1. To create main.txt on main branch and to stage and commit it

Created and navigated to directory Assignment-4

```shell
cd D:\intellipaat\Devops\Assignments

git switch main

mkdir Assignment-4
cd Assignment-4
```

```
PS C:\Users\moham> cd D:\intellipaat\Devops\Assignments
PS D:\intellipaat\Devops\Assignments> git swtich
```

```
PS D:\intellipaat\Devops\Assignments> mkdir Assignment-4


    Directory: D:\intellipaat\Devops\Assignments


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-11-2026     11:42                Assignment-4


PS D:\intellipaat\Devops\Assignments> cd Assignment-4
PS D:\intellipaat\Devops\Assignments\Assignment-4>
```

Created main.txt

```none
"This is the main file for Assignment 4." | Set-Content main.txt
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> "This is the main file for Assignment 4." | Set-Content main.txt
```

To Stage, commit and push:

```shell
git add main.txt
git commit -m "Assignment 4: Add main.txt"
git push origin main
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git  add main.txt
PS D:\intellipaat\Devops\Assignments\Assignment-4> git commit -m "Assignment 4: Add main.txt"
[main 0f0ae4f] Assignment 4: Add main.txt
 1 file changed, 1 insertion(+)
 create mode 100644 Assignment-4/main.txt
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 396 bytes | 396.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mg-ishaque/DevOps-Assignments.git
   ac73385..0f0ae4f  main -> main
```

2. To Create three Git branches

Created the three branches

None

```
git branch public1
git branch public2
git branch private3
```

Verified branches

None

```
git branch
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git branch
* main
  private3
  public1
  public2
PS D:\intellipaat\Devops\Assignments\Assignment-4>
```

Pushed branches to GitHub

None

```
git push -u origin public1
git push -u origin public2
git push -u origin private3
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push -u origin public1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'public1' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/DevOps-Assignments/pull/new/public1
remote:
To https://github.com/mg-ishaque/DevOps-Assignments.git
 * [new branch]      public1 -> public1
branch 'public1' set up to track 'origin/public1'.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push -u origin public2
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'public2' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/DevOps-Assignments/pull/new/public2
remote:
To https://github.com/mg-ishaque/DevOps-Assignments.git
 * [new branch]      public2 -> public2
branch 'public2' set up to track 'origin/public2'.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push -u origin private3
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'private3' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/DevOps-Assignments/pull/new/private3
remote:
To https://github.com/mg-ishaque/DevOps-Assignments.git
 * [new branch]      private3 -> private3
branch 'private3' set up to track 'origin/private3'.
```

3. To Create `public1.txt` on the `public1` branch

Switched to the public1 branch

Shell

```
git switch public1
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch public1
Switched to branch 'public1'
Your branch is up to date with 'origin/public1'.
```

To Create public1.txt

Shell

```
"This file belongs to the public1 branch." | Set-Content public1.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> "This fie
l belongs to the public1 branch." | Set-Content public1.txt
```

To stage and commit

Shell

```
git add public1.txt
git commit -m "Assignment 4: Add public1.txt on public1 branch"
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git add public1.txt
PS D:\intellipaat\Devops\Assignments\Assignment-4> git commit -m "Assignment 4: Add public1.txt
on public1 branch"
[public1 65aabf3] Assignment 4: Add public1.txt on public1 branch
 1 file changed, 1 insertion(+)
 create mode 100644 Assignment-4/public1.txt
```

Push changes to GitHub

```
None
git push -u origin public1
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push -u origin public1
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 426 bytes | 426.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mg-ishaque/DevOps-Assignments.git
   0f0ae4f..65aabf3  public1 -> public1
branch 'public1' set up to track 'origin/public1'.
PS D:\intellipaat\Devops\Assignments\Assignment-4>
```

4. To merge `public1` into `main`

To switch to the main branch

```
Shell
git switch main
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

To merge public1 into main

```
Shell
git merge public1
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge public1
Updating 0f0ae4f..65aabf3
Fast-forward
 Assignment-4/public1.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Assignment-4/public1.txt
```

To push main to GitHub

```
Shell
git push origin main
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mg-ishaque/DevOps-Assignments.git
   0f0ae4f..65aabf3  main -> main
```

5. **To Merge `public2` into `main`**

To merge `public2` into `main`

```
Shell
git merge public2
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge public2
Already up to date.
```

To push to GitHub

```
Shell
git push origin main
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin main
Everything up-to-date
```

6. **To edit `main.txt` on the `private3` branch, stage and commit the changes.**

To switch to the private3 branch

```
Shell
git switch private3
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch private3
Switched to branch 'private3'
Your branch is up to date with 'origin/private3'.
```

To edit main.txt and appended a new line:

```
Shell
"Updated main.txt from private3 branch - Assignment 4 Task 6." |
Add-Content main.txt
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> "Updated main.txt from
private3 branch - Assignment 4 Task 6." | Add-Content main.txt
```

To stage and commit

```shell
git add main.txt
git commit -m "Assignment 4: Update main.txt on private3 branch"
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git add main.txt
PS D:\intellipaat\Devops\Assignments\Assignment-4> git commit -m "Assignment 4: Update main.txt on private3 branch"
[private3 12d29cf] Assignment 4: Update main.txt on private3 branch
 1 file changed, 1 insertion(+)
```

Push to GitHub

```shell
git push -u origin private3
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push -u origin private3
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 438 bytes | 438.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mg-ishaque/DevOps-Assignments.git
   0f0ae4f..12d29cf  private3 -> private3
branch 'private3' set up to track 'origin/private3'.
```

7. To update `public1` and `public2` with changes from `private3`

To switch to public1

```shell
git switch public1
```

To merge private3 into public1

```shell
git merge private3
```

To Push public1 to GitHub

```shell
git push origin public1
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch public1
Switched to branch 'public1'
Your branch is up to date with 'origin/public1'.
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge private3
Merge made by the 'ort' strategy.
 Assignment-4/main.txt | 1 +
 1 file changed, 1 insertion(+)
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin public1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 399 bytes | 399.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mg-ishaque/DevOps-Assignments.git
   65aabf3..22dc597  public1 -> public1
```

To switch to public2

```shell
git switch public2
```

To merge private3 into public2

```shell
git merge private3
```

To push public2 to GitHub

```shell
git push origin public2
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch public2
M       Assignment-4/main.txt
Switched to branch 'public2'
Your branch is up to date with 'origin/public2'.
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge private3
Updating 0f0ae4f..12d29cf
Fast-forward
 Assignment-4/main.txt | 1 +
 1 file changed, 1 insertion(+)
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin public2
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mg-ishaque/DevOps-Assignments.git
   0f0ae4f..12d29cf  public2 -> public2
```

8. To Update `main` with the new code from `private3`

Switch to the main branch and merge private3 into main

```shell
git switch main
git merge private3
git push origin main
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge private3
Merge made by the 'ort' strategy.
 Assignment-4/main.txt | 1 +
 1 file changed, 1 insertion(+)
PS D:\intellipaat\Devops\Assignments\Assignment-4> Get-Content main.txt
This is the main file for Assignment 4.
Updated main.txt from private3 branch - Assignment 4 Task 6.
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 390 bytes | 390.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mg-ishaque/DevOps-Assignments.git
   65aabf3..123112a  main -> main
```

9. To update all the code on the `private3` branch

To switch to the private3 branch and Merge main into private3

```shell
git switch private3
git merge main
git push origin private3
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-4> git switch private3
Switched to branch 'private3'
Your branch is up to date with 'origin/private3'.
PS D:\intellipaat\Devops\Assignments\Assignment-4> git merge main
Updating 12d29cf..123112a
Fast-forward
 Assignment-4/public1.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Assignment-4/public1.txt
PS D:\intellipaat\Devops\Assignments\Assignment-4> git push origin private3
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mg-ishaque/DevOps-Assignments.git
   12d29cf..123112a  private3 -> private3
PS D:\intellipaat\Devops\Assignments\Assignment-4>
```

Summary and Observation

In this assignment, I learned how Git branching and merging work when multiple developers are working on the same project. The main objective was to understand how developers can work independently on different branches and later merge their changes into the main branch.

I created three branches (`public1`, `public2`, and `private3`) and performed file modifications, commits, and merges between them.

Key Observations:

1. **Branch Independence:** Changes made in one branch do not automatically affect other branches. Each branch has its own commit history until the changes are merged.
2. **Merging:** When I merged `public1` into `main`, the main branch received the changes, but `public2` and `private3` remained unchanged.
3. **Branch Synchronization:** To get the latest changes in other branches, I had to merge the updated code into each branch separately.
4. **Commit History:** Git keeps track of changes through commits, which helps in managing and combining code from different branches.
5. **Already Up to Date:** I also observed that Git displays this message when the current branch already contains all the commits from the branch being merged.

Conclusion:

This assignment helped me understand how branching and merging are used in real DevOps projects where multiple developers work on the same codebase. One important thing I learned is that merging changes into `main` does not automatically update the other branches. Each branch needs to be updated separately to get the latest code. This helps developers work independently while still being able to combine their changes when needed.